

Towards causally aware sensitivity analysis?

Motivation

The correlation pitfalls

Sobol indices and Shapley values are correlation based

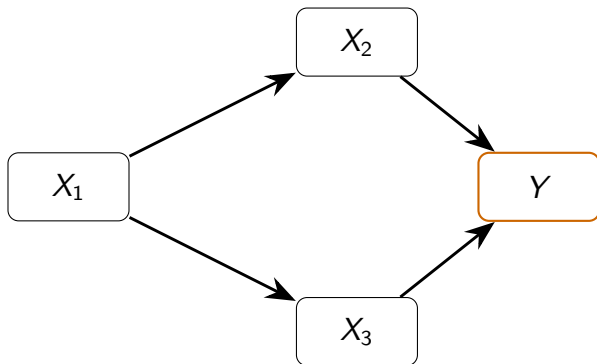
- ▶ A feature can look important simply because it is a proxy for the true cause
- ▶ Marginal and conditional Shapley values cannot separate a feature's own effect from what it only mediates.

Motivation

The correlation pitfalls

- ▶ Toy structural model: X_1 drives X_2 and X_3 , which both feed Y directly
- ▶ X_1 itself has **no direct edge into** Y .

Motivation



See `Lecture2a.ipynb`

Motivation

Towards actionable explanations

- ▶ Aim : go beyond observational setting
- ▶ Tells you what to change
- ▶ Gives you a valid guarantee about what will happen if you do.
That's two separate requirements, and most explanation methods satisfy at most one:

Motivation

Mutability

- ▶ The explanation must point to a variable an agent can actually manipulate.
- ▶ "Age" can be causally important without being actionable
- ▶ Recourse requires identifying feasible changes, not just important ones

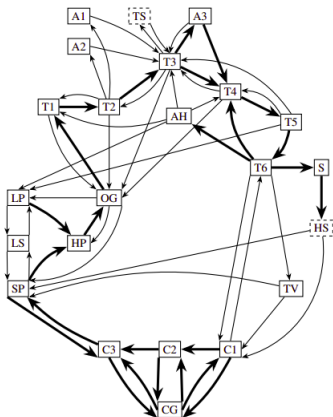
Motivation

Quantifying an action effect

- ▶ If a feature is flagged as important, acting on it should have an important effect
- ▶ This is not measured with correlation
- ▶ Towards **causally aware** importance measures?

Why causality?

A motivating dataset from the European Spallation Source (ESS)



Motivation

- ▶ The data was generated by the ACCP system (a refrigeration system),
- ▶ A number of process variables (PVs) describing the state of the ACCP were measured over time.
- ▶ Observations from three separate time intervals are available (Period 1, 2, and 3).
- ▶ Each PV belongs to a subsystem, and each subsystem is a physical component of the ACCP system (such as a compressor).

Motivation

Index	Description	No. of PVs	Index	Description	No. of PVs
T1	Turbine 1	5	C1	cold Compressor 1	5
T2	Turbine 2	5	C2	cold Compressor 2	6
T3	Turbine 3	8	C3	cold Compressor 3	6
T4	Turbine 4	5	CG	Compressor General	11
T5	Turbine 5	6	SP	Sub-atm. Prs. compressor	17
T6	Turbine 6	8	SPWE	SP line Warm End	2
TG	Turbine General	1	LP	Low Pressure compressor	16
A1	Adsorber 1 80 K	3	LPWE	LP line Warm End	2
A2	Adsorber 2 80 K	4	LS	LP+SP compressor	8
A3	Adsorber 3 20 K	4	MPWE	MP line Warm End	3
HX1	Heat eXchanger 1	2	HP	High Pressure compressor	22
HX2	Heat eXchanger 2	9	HPWE	HP line Warm End	5
HX3	Heat eXchanger 3	2	OG	Oil and Gas removal	15
HX4	Heat eXchanger 4	3	AH	Ambient Heater	5
HX5	Heat eXchanger 5	2	D	Dewar	9
HX6	Heat eXchanger 6	5	S	Subcooler	7
TV	Test Vessel	8	TS	Thermal Shield*	8
HS	Helium Supply*	6			

Motivation

Practical questions

- ▶ Can we intervene to control the output of the system?
- ▶ Can we prevent derives/anomalies?
- ▶ Can we identify the root causes of such derives/anomalies and derive a strategy of remediation

Causal modeling : a gold road to answer such questions?

What is causality?

Causal and non causal questions

- ▶ Non-causal: investigate purely statistical associations
- ▶ Causal effects : Estimating causal effects (of hypothetical interventions) using the causal graph
- ▶ Counterfactuals : Such as attribution of observed events or mediation analysis

Caveat : the graph **could be unknown** and in this case we have to estimate it!

First scientific question : causal discovery

What about the case where the DAG is unknown?

- ▶ In general, causal discovery from observational data is not possible.
- ▶ But it is possible under additional assumptions.

Second scientific question : causal inference, effect of an intervention

Causal quantities

- ▶ Assume now DAG known or estimated
- ▶ We cannot quantify the output of an intervention for a given individual
- ▶ A causal quantity

$$ATE = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

Second scientific question : causal inference, effect of an intervention

Scientific questions

- ▶ Practical estimation of the causal effects from (observational) data?
- ▶ Do calculus, identifiability questions

See `Lecture2b.ipynb`